

Paulette Suro

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EDUCATION

San Diego State University

San Diego, CA

Bachelor of Science in Mechanical Engineering, Bioengineering Emphasis | August 2021 - May 2025

Master of Science in Bioengineering | August 2025 - July 2026 | 3.33 GPA

Thesis “Redox Amplification in Glassy Carbon Electrodes for Glucose Sensing through Sweat.” Designed, fabricated, iteratively optimized, and characterized a MEMS electrochemical sensor through the full product development cycle using semiconductor microfabrication for non-invasive monitoring. Developed design requirements and maintained traceable design change documentation across three design iterations mirroring design control/DHF documentation practices. Performed design verification using CV/ EIS in vitro testing, applying regression analysis to confirm electrode behavior.

SKILLS

Software Tools: CAD (SOLIDWORKS, AutoCAD, CoventorWare), MATLAB, Arduino, Python, COMSOL, Gamry Analyst.

Skills: English and Spanish, Presentation & Technical Communication, R&D, Testing, Data Analysis, Project Leadership, MEMS Fabrication, Mask Layout, Mechanical Design, Hardware Integration, Electronics Testing, Rapid Prototyping.

WORK EXPERIENCE

Lab Manager, SDSU Research Foundation - NanoFAB.SDSU

San Diego, CA | May 2023 - July 2026

- Served as Mask Layout Lead, compiled researchers' designs into full mask layouts and conducted design reviews to reduce fabrication rework and accelerate project timelines; developed SOPs and engineering documentation for the mask design and ordering process, which reduced processing time from a week to two days.
- Trained researchers in MEMS fabrication through hands-on instruction in photolithography, thin-film deposition, etching, CAD mask design, and cleanroom processing, collaborating cross-functionally to drive process improvement.

Firmware Engineering Intern, Universal Electronics

Poway, CA | May 2025 - August 2025

- Automated relay validation across 20 embedded firmware safety test sequences using Python, Z-Wave, and Zigbee, improving test repeatability across multiple hardware platforms and reducing manual testing time by an hour per run.

Senior Design Project Suspension Lead, SDSU Senior Design - SAE Aztec Baja

San Diego, CA | August 2024 - May 2025

- Designed, machined, and integrated a double-wishbone front suspension subsystem using SOLIDWORKS CAD, implementing a user-controlled shock stiffness adjustment and a 15.8% turn radius reduction to improve handling.
- Validated electronic shock control systems through wiring, soldering, and test stand testing; designed and 3D printed welding fixtures and tooling to support suspension fabrication, applying risk management to fabrication decisions.
- Presented suspension design and performance results at the Arizona SAE Baja competition, won 2nd in sled pull.

Hardware Engineering Intern, Universal Electronics

Poway, CA | May 2024 - August 2024

- Designed and built an automated hardware validation test equipment platform to evaluate thermostat reboot reliability under fluctuating voltage using oscilloscopes and control scripts; ran a thermal distribution study (thermistors, MATLAB, DAQ) that identified uneven heat distribution and proposed soak-time adjustments (DOE).

Robotics Instructor, Smart Mind Robotics

La Mesa, CA | October 2020 - April 2023

- Taught robotics principles and programming to students in grades 3-8, adapting lessons across learning styles/levels.

SELECTED ENGINEERING PROJECTS

Bioelectronic Device-Epilepsy Treatment: Designed and fabricated a closed-loop neurostimulation medical device integrating MEMS fabrication, electrochemical sensing, hardware prototyping, and system integration for responsive stimulation.

Bioelectronic Device-Diabetes Treatment: Designed, fabricated, and characterized an implantable MEMS cuff electrode medical device for vagus nerve stimulation to enhance insulin sensitivity, supporting the development of a bioelectronic alternative to traditional diabetes therapies; won an undergraduate research excellence award at the SDSU Spring Symposium.

3D Printed Transtibial Prosthetic: Developed a seahorse tail-inspired prosthetic attachment via 3D resin printing to provide passive ankle articulation and improve gait stability on uneven terrain without costly electronics or hydraulics.

LEADERSHIP EXPERIENCE & ACTIVITIES

Biomedical Engineering Society (BMES), Vice-President

San Diego, CA | June 2024 - May 2026

- Led professional development initiatives by planning workshops, guest speakers, and industry/research lab tours.
- Founded the project subteam, achieved BMES national membership for SDSU, and grew membership from 11 to 26.

American Society of Mechanical Engineers (ASME), President, Treasurer

San Diego, CA | June 2023 - May 2025

- Led meetings and grew paid membership by **165%** through active outreach and value-driven programming.
- As treasurer, secured a \$17,000 grant through a funding proposal and managed budgeting and reimbursements.